The risk of nuclear conflict has re-emerged as a pressing concern in contemporary international relations. The New START treaty between the United States and Russia expired on 5th February 2026 without a successor agreement (Liang, 2026), and all nine nuclear-armed states are actively modernizing their arsenals with some states also increasing warhead numbers (Stockholm International Peace Research Institute, 2025; Fayet, 2026). Against this backdrop, a re-examination of the potential consequences of nuclear conflict is warranted. These were first systematically explored during the 1980s, when early modeling studies showed that post-nuclear firestorms could inject black carbon into the atmosphere, triggering sustained surface cooling, a phenomenon termed nuclear winter (Crutzen & Birks, 1982; Aleksandrov & Stenchikov, 1983; Turco et al., 1983). With the end of the Cold War, research expanded to include more limited regional exchange scenarios, and modern coupled climate models have since confirmed and in some cases strengthened early projections, demonstrating that even a regional nuclear exchange could produce surface cooling of a magnitude not seen since the last Ice Age (Robock et al., 2007a; Mills et al., 2014; Coupe et al., 2019; Toon et al., 2019). Surface cooling would substantially reduce evapotranspiration and, consequently, precipitation, weakening the hydrological cycle as a whole (Robock et al., 2007b). Nuclear winter conditions would alter not only how much freshwater is available, but also when it becomes available — a distinction that is critical for human water security. Among the processes governing the timing of freshwater availability, seasonal snow storage plays a particularly important role. Acting as a natural reservoir, it captures winter precipitation in solid form and releases it as meltwater during warmer months, when irrigation demand and evapotranspirative losses are higher (Hayhoe et al., 2004; Barnett et al., 2005, 2008). This buffering function supports freshwater supplies for approximately 1.9 billion people worldwide and is critical for irrigation systems across High Mountain Asia, Central Asia, the western United States, and the southern Andes, where agricultural production depends heavily on snowmelt-fed runoff (Immerzeel et al., 2020; Qin et al., 2020). While volcanic eruptions offer a partial analogue and have been shown to alter river discharge patterns (Iles & Hegerl, 2015; Kim et al., 2025), the specific implications of nuclear winter for snow accumulation, melt timing, and the resulting seasonality of runoff remain poorly constrained. The global temperature reduction following the soot injection affects snow dynamics through three distinct mechanisms: the emergence of new seasonality in previously snow-free regions, the prolongation of existing snow seasons, and the continuous multi-year accumulation in high-latitude regions.The first mechanism is most directly visible in the rapid expansion of snow cover extent in Year 1 and 2: as winter temperatures are pushed below the freezing threshold in regions that rarely experience sub-zero conditions, precipitation

shifts from rain to snow, generating entirely new seasonal snow signals as seen in Cluster 1. The spatial footprint of this response reflects the southward displacement of the winter freezing isotherm under nuclear winter conditions, a pattern that scales with the magnitude of soot injection (Coupe et al., 2019). Observational evidence from volcanic eruptions supports the plausibility of this mechanism: snowfall has been documented in the Middle East in the winter of 1172/73 CE following a major medieval eruption in 1170/71 CE, and in the winter of 1991/92 CE following the Pinatubo eruption, though a causal link for the medieval case remains inconclusive (Guillet et al., 2025). Because this new seasonality is directly tied to the temperature threshold being crossed, both the recovery of Cluster 1 and the global snow cover extent anomaly track the temperature recovery more closely than mean snow storage.The second mechanism, prolongation of existing snow seasons, is reflected in the timing of the global mean storage anomaly and snow cover extent anomaly, which peak in May or June rather than in winter. This indicates that snow persists into months that would normally see rapid melting, driven by the stronger negative temperature anomalies over land during summer relative to winter (Coupe et al., 2019). Cluster 2 illustrates this mechanism particularly clearly, as the period with seasonal snow cover extends substantially relative to the control. However, the concurrent reduction in total precipitation causes annual snow storage maxima to gradually fall below control levels. The third mechanism, continuous multi-year accumulation, is reflected by Cluster 3 in high-latitude regions, where temperature reductions are large enough to suppress the melt season almost completely, allowing snowpacks to build up continuously in the first years after the soot injection. Even as global temperatures begin to recover, they remain below the melt threshold in these regions, which explains why the maximum of the global snow storage anomaly is delayed relative to the maximum of the temperature reduction. Subsequent recovery is comparably slow, as it requires substantial latent heat input to melt these accumulated snowpacks. The spatial pattern of these three regimes broadly corresponds to the reduction in growing season length projected by Coupe et al. (2019) and the increase in maximum frost depth described by Lamilla Cuellar et al. (2026), with Cluster 3 regions facing near-complete elimination of the growing season in the first years following the soot injection and the largest frost depth increases, while Cluster 2 regions experience growing season reductions of 50-150 days. This correspondence across independent analyses underlines that the snow regimes identified here are part of a coherent pattern of regional climate disruption. It should be noted, however, that while the land-surface model underlying these frost depth estimates (Oleson et al., 2010) accounts for the insulating effect of snow on soil heat loss (Zhang, 2005), the reliability of this representation under the deep, multi-year snowpacks

simulated here remains uncertain, leaving open the possibility that the frost depth increases reported by Lamilla Cuellar et al. (2026) are somewhat overestimated in these regions. A further notable feature is that snow storage anomalies do not return to control levels within the 15-year simulation period across all scenarios, including the 16 Tg case. This persistent anomaly reflects the equally persistent temperature depression reported by Coupe et al. (2019) and Toon et al. (2019), which itself does not return to pre-war levels within the simulation period. As Harrison et al. (2022) demonstrate, even relatively moderate soot injections trigger a hysteresis response in the ocean that persists well beyond the atmospheric forcing, with sea ice expansion and enhanced vertical mixing maintaining a cooled ocean state for decades, providing a physical basis for the incomplete recovery observed here. The persistence of large snowpacks in high-latitude regions such as those represented by Cluster 3 and 4 would further amplify this signal through the snow-albedo feedback: additional snow cover reflects a greater fraction of incoming solar radiation, suppressing surface warming and delaying the recovery of surface temperatures. Robock (1984) demonstrated this mechanism using an energy balance model, showing that snow and ice feedbacks can substantially amplify and prolong the climatic response to nuclear winter forcing beyond what atmospheric processes alone would predict. Not all snow storage anomalies in this study are driven by the same mechanism, nor are they uniformly positive. While the increases observed in Clusters 1 and 3 are primarily driven by cooling-induced changes in snowfall and melt, this temperature-driven mechanism is progressively counteracted by the concurrent decline in precipitation in Cluster 2, eventually driving its annual maxima below control levels. In Cluster 4, where temperatures already remain below the snowfall threshold under both control and scenario conditions, this temperature-driven mechanism is absent altogether, leaving the decline in precipitation as the dominant influence; this manifests as pronounced and sustained negative snow storage anomalies in parts of the cluster, particularly Greenland, although the cluster-level signal remains comparatively weak because of strong spatial heterogeneity. This divergence illustrates that the response to nuclear winter forcing is not governed by a single mechanism, even within terrestrial snow storage alone. A similar divergence is evident beyond terrestrial snow. While this study identifies a robust increase in terrestrial snow storage across the Northern Hemisphere, cryospheric responses to nuclear winter forcing are not uniformly positive globally either. Coupe et al. (2023) demonstrate that Antarctic sea ice undergoes a substantial decline in the largest nuclear war scenario despite global cooling, driven by wind-stress-induced ocean circulation changes. Together, these findings suggest that nuclear winter reorganises the cryosphere in spatially complex ways rather than producing a uniform expansion of snow and

ice. While cooling promotes snow accumulation across large parts of the Northern Hemisphere, precipitation-driven reductions in snow storage remain possible in permanently frozen regions, and Antarctic sea ice may even decline despite lower global temperatures.The reorganization of precipitation from rain to snow under nuclear winter conditions fundamentally reshapes the seasonal distribution of runoff across most basins. First, the extended period of sub-zero temperatures prolongs the phase during which little to no runoff is generated from either rainfall or snowmelt. As precipitation falls predominantly as snow and temperatures remain too low to trigger melting, this water is effectively stored in the snowpack rather than contributing to discharge. River discharge during this phase would be sustained primarily by groundwater baseflow, which may itself be reduced under freezing conditions. Wang (2019) demonstrate that low temperatures can decrease aquifer hydraulic conductivity, thereby constraining groundwater contributions to streamflow even where groundwater storage remains available. In addition, groundwater recharge may be both reduced and increasingly confined to warmer periods. Prolonged soil freezing and the delayed onset of snowmelt could postpone infiltration. At the same time, the general reduction in total precipitation limits the overall volume available for recharge. This potentially increases the risk of long-term groundwater depletion, although quantifying this would require dedicated groundwater modelling beyond the scope of this study. These hydrological changes have direct implications for freshwater availability. Under nuclear winter conditions, underground water supply infrastructure is simultaneously at risk of frost damage (Lamilla Cuellar et al., 2026), thereby increasing dependence on surface water resources precisely during periods of minimum runoff availability. Second, the extended snow season allows for prolonged accumulation, resulting largely in greater snowpack volumes. As melt remains confined to a broadly similar seasonal time window, these larger volumes generate substantially higher discharge peaks, with potential implications for flood risk, as evident in the Rhine and Tigris-Euphrates basins. This effect is further amplified by the higher runoff coefficient of snowmelt relative to rainfall: meltwater release occurs during periods of naturally low evapotranspiration, allowing a larger fraction to be converted directly into discharge (Berghuijs et al., 2014; Jenicek & Ledvinka, 2020), an effect likely reinforced by the broad suppression of evapotranspiration under nuclear winter conditions (Robock et al., 2007b). In some regions, delayed melt may additionally increase the potential for overlap with summer rainfall events, although this effect appears limited given the strongly weakened monsoon systems across Asia, North America, and Africa under nuclear winter conditions (Robock et al., 2007b; Coupe et al., 2019).Third, not all basins follow this pattern. In the Ob and Yellow River basins, annual snowmelt anomalies are negative, primarily reflecting the general

reduction in precipitation and thus in snowfall accumulation. In the Ob, however, the accumulation mechanism of cluster 3 operates: temperatures remain too low for complete seasonal melting, causing a growing fraction of the snowpack to persist year-round, preventing it from contributing to discharge as snowmelt. This has particularly severe consequences because the Ob's hydrology is fundamentally structured around its spring flood pulse, which accounts for 70-80% of total annual runoff and derives approximately half of its volume from snowmelt (Papina et al., 2023). Under scenario conditions, snowmelt contributions to river discharge are reduced by up to 55%, directly undermining this flood pulse, with potential implications for water availability and freshwater ecosystems across this high-latitude region.Taken together, these hydrological responses are broadly consistent with previous evidence on streamflow responses to stratospheric aerosol forcing from volcanic eruptions. Kim et al. (2025) combined global climate model simulations with a statistical streamflow model applied to 7,886 gauging stations worldwide to investigate changes in seasonal peak discharges with a 2-year return interval. For the Pinatubo eruption whose symmetric aerosol distribution most closely resembles nuclear winter forcing they find a widespread reduction in the frequency of winter (DJF) peak discharge events across the Northern Hemisphere, broadly consistent with the general suppression of discharge identified here. Conversely, both the frequency and magnitude of peak discharges increase during spring and early summer (MAM and JJA) in arid and temperate regions, consistent with the elevated discharge peaks identified for the Tigris-Euphrates and Rhine basins. The agreement across both the seasonal timing and regional patterns of discharge responses suggests that the hydrological changes under nuclear winter conditions represent an intensified extension of processes already observed following major volcanic forcing. The broader implication is that nuclear winter would not simply reduce freshwater availability, but fundamentally alter its seasonal distribution, increasing the potential for both hydrological droughts and flood events across snow-influenced river basins. The snowpack anomalies identified in this study would have consequences extending well beyond their hydrological significance, most acutely in Cluster 3 regions, primarily the high-latitude regions of Russia, Scandinavia, and Canada, where multi-year snowpacks develop that reach exceptional depths. To translate snow water equivalent into snow depth, a representative bulk density of 0.3-0.4 g cm^{-3} can be assumed for settled to wind-compacted snowpacks. This range is consistent with physically based density evolution from seasonal snow to firn (Cuffey & Paterson, 2010) and with observed snowpack properties in tundra and taiga environments (Sturm et al., 2010). The median snow storage reaches approximately 1000 mm SWE after 5-6 years, corresponding to snow depths on the order of 2.5-3.3 m under these assumptions. Upper

quantiles of SWE yield substantially greater depths, reaching approximately 6-8 m in the most snow-accumulating locations. Such depths would challenge infrastructure systems designed to cope with seasonal snow accumulation but not with persistent multi-year snowpacks. Transport would be severely disrupted, with roads and railways buried under snow and airport operations greatly constrained. Maintaining transport infrastructure would require substantial effort and could impair the supply of food, fuel, and medical resources to affected populations. Heavy snow loads would additionally increase the risk of falling trees damaging overhead power lines, potentially causing widespread power outages and compounding energy insecurity at a time when heating demands are substantially elevated. The accumulating snowpacks would furthermore place exceptional structural loads on buildings, increasing the risk of roof damage or collapse. Overall, these compounding pressures suggest that extreme multi-year snowpacks would represent not only a hydrological anomaly, but also a broader challenge to societal resilience in affected regions. Temperature-index approaches implicitly assume a relatively stable relationship between air temperature and available melt energy. Even under present day conditions, degree-day factors vary considerably reflecting differences in solar radiation, albedo, snow density, and topographic factors such as slope, aspect, and elevation (Hock, 2003). In high-latitude regions where temperatures remain below the melt threshold year-round, the approach is not applicable at all, as it attributes mass loss solely to above-freezing temperatures; such regions were therefore excluded from the hydrological discussion. Under nuclear winter conditions, the core assumption is further challenged by a temporal decoupling of radiation and temperature: reductions in incoming solar radiation precede the surface temperature minimum, while temperature recovery is delayed by ocean thermal inertia (Coupe et al., 2019; C. S. Harrison et al., 2022). As a result, the degree-day factor is likely overestimated during the early years of the simulation, when radiation is already strongly reduced but temperatures have not yet reached their minimum, and potentially underestimated during the recovery phase, when radiation recovers faster than surface temperature. While the degree-day factor applied here falls within the range reported in the literature (Massmann, 2019), this systematic temporal bias should be considered when interpreting melt-driven discharges. Additionally, the discharge estimates represent the total liquid water input to each catchment from snowmelt and rainfall, without accounting for catchment-scale partitioning into infiltration, soil moisture storage, groundwater recharge, or evapotranspiration. They therefore represent an upper bound of potential surface water availability rather than actual river discharge. This means that the gap in water availability between scenario and control is likely larger than the raw discharge values suggest, as evapotranspiration losses are smaller under nuclear winter conditions. The monthly

temporal resolution of the forcing data limits the representation of sub-monthly snowmelt dynamics and precludes the explicit simulation of rain-on-snow events. Consequently, short-term discharge peaks and flood-generating processes cannot be assessed. Daily forcing data would provide a more realistic representation of melt onset and intensity, rain-on-snow occurrence, and extreme discharge events. However, the monthly resolution is sufficient to analyse the large-scale seasonal changes in snow storage and runoff timing that are the focus of this study.